Equity Return Prediction from Fundamentals Data: Applying XGBoost to Changing Market Regimes

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ABSTRACT

A substantial literature links accounting fundamentals to future equity returns. Foundational work by Ou and Penman (1989), Piotroski (2000), and Novy-Marx (2013) demonstrates that financial-statement signals contain information beyond what is reflected in market prices. More recent machine-learning studies—notably Gu, Kelly, and Xiu (2020) and Freyberger, Neuhierl, and Weber (2020)—apply non-linear methods to large characteristic sets and report improvements in rank information coefficients. However, these studies share important limitations: results are typically framed as factor risk-premia rather than implementable return metrics; safeguards against survivorship bias are inconsistently applied; evaluation periods are short and do not span the full range of market regimes; and the gap between reported performance and a practically executable strategy is rarely closed.

This paper presents an end-to-end quantitative framework that addresses these limitations directly. Using a point-in-time fundamental database covering over 5,700 U.S. equities, we report out-of-sample results on a strict holdout test period from 2012 to 2023—spanning the post-crisis bull market, the COVID-19 shock, and the 2022 rate cycle. An XGBoost gradient-boosted tree model is trained with a custom three-block time-series cross-validation (TSCV) design. To better address the well-known challenge of changing market regimes, the validation block is placed *prior* to the training block, hence significantly shortening the gap between the training and test periods. This design allows the model to adapt more fluidly to evolving market regimes by minimizing the temporal gap between the training window and the evaluation period. Applied to a long-only top-30 strategy, the framework produces annualized returns of approximately 14% versus 7% for the equal-weighted universe benchmark over the test period, with the strategy outperforming the benchmark in 61% of individual monthly periods and in over 77% of rolling 12-period windows. The predictive signal is broad rather than concentrated: performance remains strong when expanding the portfolio to the top 100 highest-ranked stocks, and the lowest-ranked decile consistently underperforms the benchmark—both outcomes suggesting genuine cross-sectional discrimination rather than overfitting to a narrow selection criterion.

Keywords: equity return prediction, fundamental analysis, XGBoost, gradient boosting, time-series cross-validation, post-earnings announcement drift, walk-forward evaluation, survivorship bias, quantitative investing, machine learning in finance

1. INTRODUCTION

The relationship between accounting fundamentals and subsequent stock returns has attracted sustained attention in academic finance and quantitative practice. The core hypothesis—that markets do not

immediately incorporate all the information embedded in financial statements—is supported by a long sequence of empirical findings, from the earnings post-announcement drift documented by Ball and Brown (1968) to the profitability premium of Novy-Marx (2013). The practical question is whether this predictability can be systematically exploited in a long-only portfolio context, over a long enough evaluation window to span multiple market regimes, and without the methodological shortcuts that inflate many published results.

This paper describes a framework built around three design principles. First, the prediction problem is framed temporally: at each decision point, predict 60-trading-day forward returns using only information available at that moment. Second, the evaluation methodology replicates sequential deployment through a custom three-block walk-forward TSCV loop. Third, the primary performance metrics are strategy returns and their statistical robustness—not RMSE or Sharpe ratios that bear only an indirect relationship to portfolio profitability.

2. LITERATURE CONTEXT AND MOTIVATION

2.1 Fundamental Signals and Return Predictability

The academic literature on fundamental-based return prediction is extensive. Piotroski (2000) showed that a nine-signal financial-health score effectively separates winners from losers among high book-to-market firms. Novy-Marx (2013) documented a gross profitability premium orthogonal to standard value factors, suggesting that revenue efficiency carries independent predictive information. Sloan (1996) demonstrated that the accrual component of earnings predicts return reversals. Together, these studies establish that the information content of financial statements is non-trivial and persistent over long samples.

2.2 Machine Learning Approaches

Gu, Kelly, and Xiu (2020) benchmarked a range of models—including random forests, gradient-boosted trees, and neural networks—against linear factor models on a large panel of firm characteristics, confirming that non-linear interactions carry predictive content beyond linear combinations. Freyberger, Neuhierl, and Weber (2020) applied LASSO-based nonparametric methods to characteristic selection, confirming the robustness of a small core set of signals. These results motivate the use of gradient boosting as the modeling framework here.

2.3 Limitations of Existing Work

Despite this progress, several important limitations constrain the practical relevance of the existing literature:

1. **Performance reporting.** Most studies report factor alphas or rank ICs. These do not directly correspond to returns achievable by a long-only investor. Cumulative wealth and the probability of benchmark outperformance over realistic horizons are rarely reported.
2. **Survivorship and look-ahead bias.** Studies using non-point-in-time databases systematically overstate predictive power, especially in periods covering market dislocations.

3. **Narrow evaluation windows.** Many ML-based studies cover 10–20 years of data and do not span the full range of market regimes, including severe liquidity crises and structural shifts in market composition.

The present framework directly addresses all three limitations.

3. DATA

The study uses two primary data sources. **Equity prices** consist of daily closing prices for U.S.-listed equities, covering over 9,000 unique tickers including firms that were subsequently delisted. The inclusion of delisted firms is essential to avoid survivorship bias: excluding firms removed from indices due to poor performance would materially inflate apparent strategy returns. **Fundamental data** are drawn from the Sharadar SF1 point-in-time database, which ensures that each observation reflects only information publicly available at the time of filing, with restatements handled correctly. Key variables include net income, free cash flow, revenue, total equity, total debt, market capitalization, and enterprise value, at quarterly (ARQ) and trailing-twelve-month (TTM) frequencies.

A minimum market capitalization filter (approximately $100 million) is applied to limit exposure to micro-cap illiquidity. All criteria are applied strictly on a point-in-time basis. The model is trained on data from 1999 to 2011 (train phase) and evaluated out-of-sample on 2012 to 2023 (test phase). All results reported in this paper are from the test phase only.

4. METHODOLOGY

4.1 Prediction Target and Features

The prediction target is the log ratio of the 60-trading-day forward closing price to the current closing price. Features are constructed from the most recent point-in-time fundamental snapshot:

- **Earnings yield and cash-flow yield** — net income and free cash flow scaled by market capitalization.
- **Relative valuation signals** — firm-level earnings yield minus the contemporaneous cross-sectional and sector means.
- **Revenue efficiency** — revenue scaled by market capitalization, a proxy for the Novy-Marx (2013) profitability signal.
- **Lagged fundamental changes** — one-period lags of earnings and cash flow, capturing earnings momentum.
- **Price momentum** — 60-trading-day trailing return as a short-term control.
- **Report timing** — business days elapsed since the most recent earnings release, capturing post-announcement drift.

Continuous features are log-transformed where appropriate and z-scored within each training window. Sample weights combine a time-decay component with an absolute-return component.

4.2 Model: XGBoost

The predictive model is XGBoost (Chen and Guestrin, 2016), a gradient-boosted decision tree algorithm chosen for its performance on tabular data with non-linear feature interactions and its native support for regularization. Key hyperparameters are: learning rate $\eta = 0.02$, maximum tree depth = 3, L2 regularization $\lambda = 0.3$, and a minimum-gain penalty $\gamma = 0.1$. These parameters favour shallow, regularized trees that generalize across regimes rather than memorizing in-sample patterns.

4.3 Three-Block Walk-Forward TSCV Design

The evaluation design is the central methodological contribution of this work. For each monthly test period, three strictly non-overlapping data blocks are constructed in temporal order:

1. **Validation block.** Drawn from the period *before* the training window, restricted to the same post-announcement subset used at test time. The number of boosting rounds is selected by early stopping on this block.
2. **Training block.** A historical window ending at least 61 trading days before the test period, ensuring no realized forward returns overlap with test labels.
3. **Test block.** One forward calendar month—the strictly out-of-sample evaluation set, never seen during training or model selection.

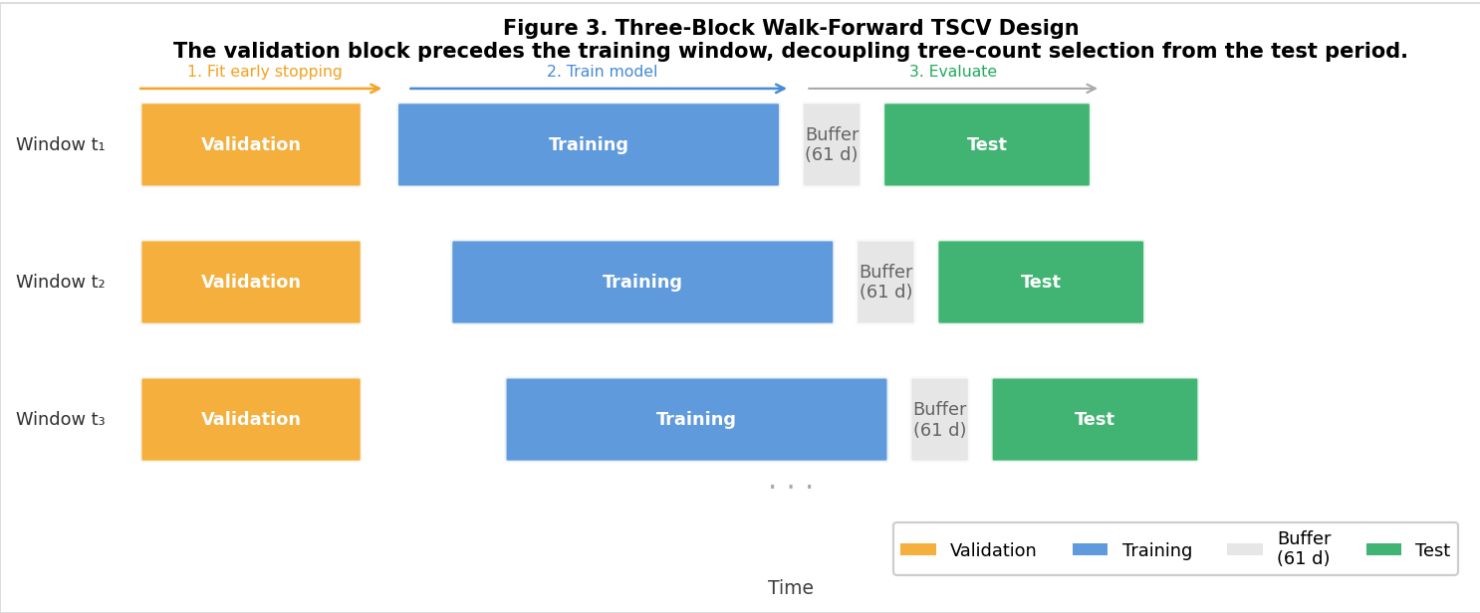

Figure 3. *Schematic of the three-block walk-forward TSCV design. For each test window, the validation block (orange) precedes the training block (blue), which in turn precedes the test block (green), separated by a buffer of at least 61 trading days. As the evaluation window advances, all three blocks shift forward in time.*

The critical innovation is the placement of the validation block. Conventional TSCV designs place the validation set between the training window and the test period. This couples model selection to market conditions immediately preceding the test period—an indirect form of temporal leakage where the selected tree count has in effect been tuned to the regime just before evaluation. Moving the validation block to the far side of the training window removes this coupling entirely, allowing the model to generalize more fluidly across regime shifts. In practice, `num_boost_rounds`—the hyperparameter controlling tree count and therefore model complexity, tuned via early stopping on the validation block—was found to vary considerably less across test periods than the model parameters learned from the training data. This stability suggests that placing the validation block earlier decouples complexity selection from near-term regime conditions: rather than calibrating the number of trees to the market environment just before the test window, the model selects a more regime-neutral level of complexity, reducing the risk of inadvertent overfitting to a transient market state.

5. RESULTS

5.1 Strategy Performance Summary

Table 1 presents the main performance metrics for the top-30 long-only strategy over the test period (January 2012 to September 2023, 141 monthly periods). At each decision point, the 30 stocks with the highest predicted 60-day return are selected from the universe of approximately 900 stocks with a recent earnings release. Positions are held for the full 60-business-day horizon before rebalancing, so portfolio turnover is modest (roughly 8–12 rebalances per year across a portfolio of a few dozen names); transaction costs are therefore not expected to materially affect the gross returns reported here.

Metric	Top-30 Strategy	Universe Benchmark
Evaluation period	January 2012 – September 2023 (141 monthly periods)	
Universe (test phase)	5,728 U.S. equities incl. delisted firms	
Annualized return	**14.2%**	6.9%
Mean 60-day period return	**3.5%**	1.9%
Std. dev. of period return	10.1%	7.8%
Prob. outperform benchmark (single period)	**61.0%**	—
Prob. outperform benchmark (rolling 12 periods, ~2 yrs)	**77.5%**	—
Mean rank information coefficient (IC)	**0.024**	—
Worst single-period return	−26.3%	−22.8%
Best single-period return	+50.1%	+26.2%

Table 1. Out-of-sample performance summary (test phase only). Returns are gross of transaction costs. The benchmark is the equal-weighted mean return of all universe stocks per period.

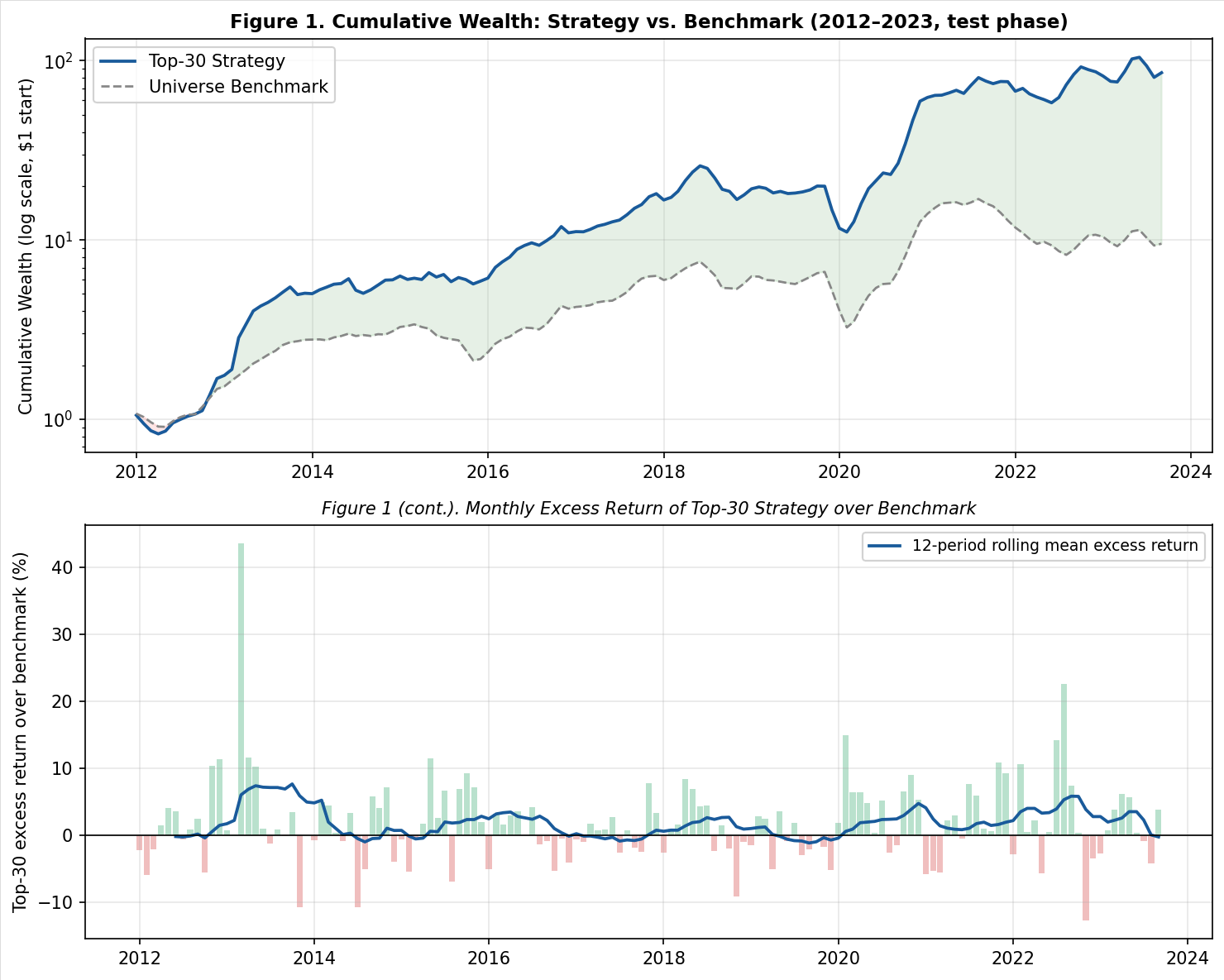

Figure 1. *Top panel: cumulative wealth of the top-30 strategy (solid blue) and the equal-weighted universe benchmark (dashed grey) on a logarithmic scale, 2012–2023 (both start at $1). Green/red shading shows periods of strategy outperformance and underperformance. Bottom panel: monthly excess return of the top-30 strategy over the benchmark (bars) and its 12-period rolling mean (solid line), illustrating persistence of the edge across sub-periods.*

The strategy more than doubles the annualized return of the benchmark (14.2% vs. 6.9%) over the 12-year test period. Outperformance is broadly distributed: the strategy beats the benchmark in 61% of individual two-month periods and in 77% of rolling two-year windows. The rolling excess return in the lower panel of Figure 1 shows that the edge is not concentrated in a narrow window—it is present across the 2012–2015 expansion, the 2018 correction, the COVID-19 shock of 2020, and the 2022 rate-driven downturn.

5.2 Signal Breadth and Cross-Sectional Discrimination

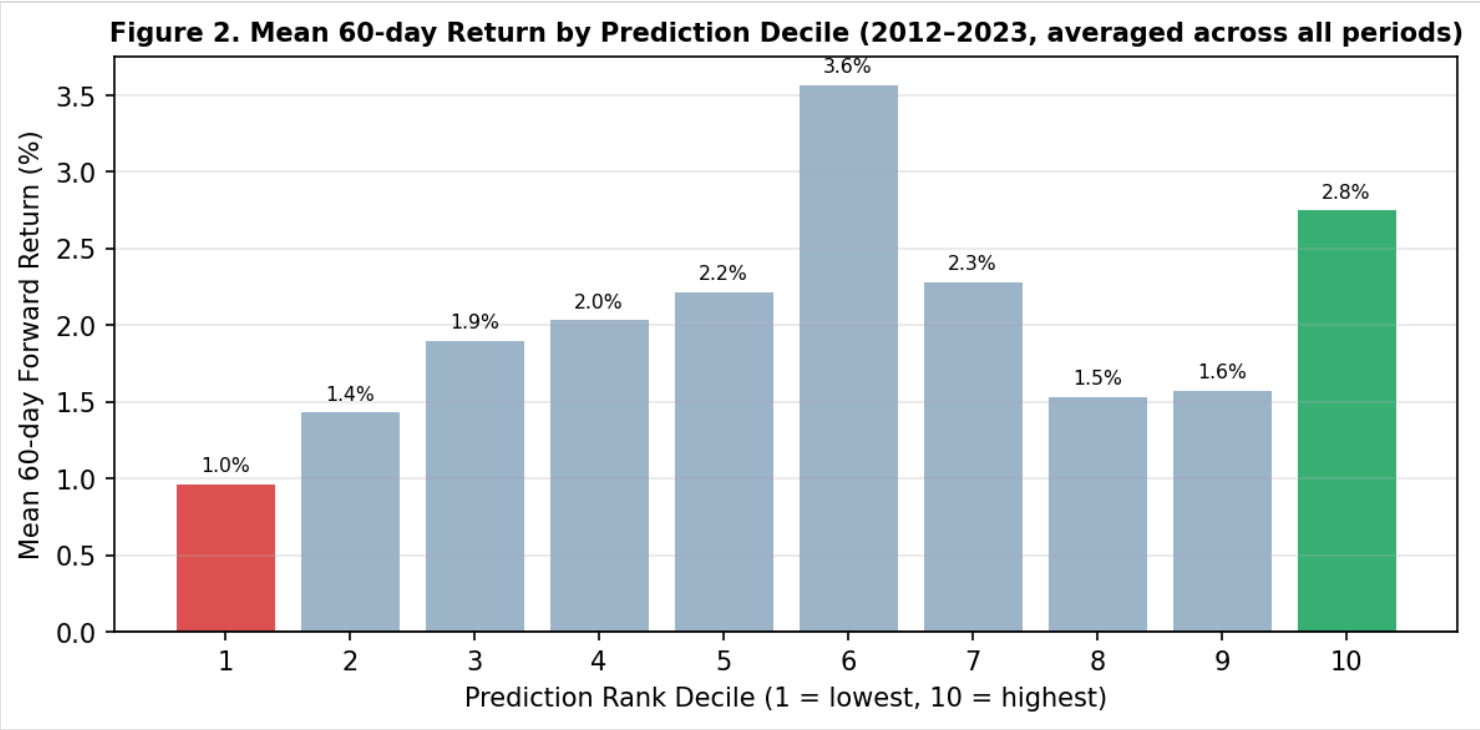

Figure 2. *Mean 60-day forward return by prediction rank decile, averaged across all 141 test-phase monthly periods. Decile 1 = lowest-ranked stocks; decile 10 = highest-ranked. The lowest decile is the worst performer, and the top decile is among the best, while middle deciles cluster around the benchmark mean.*

Two findings from Figure 2 merit particular attention. First, the lowest-ranked decile (D1) averages approximately 1% per period—consistently below the equal-weighted benchmark of 1.9%— confirming that the model discriminates meaningfully at the bottom of the distribution, not only at the top. Second, the strategy's edge is not confined to the 30 highest-ranked stocks: expanding the portfolio to the top 100 highest-ranked stocks (roughly the top decile of the approximately 900-stock eligible universe) still yields a mean per-period return of approximately 3.0%—materially above the 1.9% benchmark mean. Critically, the model was optimized solely on top-30 returns during training; the generalization to a wider selection is out-of-sample evidence of robustness rather than overfitting to a specific portfolio-size parameter.

5.3 Feature Importance and Economic Interpretation

Examining aggregate XGBoost feature importance (gain) across all test-phase monthly models, the most influential features are revenue, earnings yield (net income scaled by market cap), and their sector-relative counterparts. Lagged earnings changes and price momentum appear as secondary contributors. These rankings are broadly consistent with the published literature (Novy-Marx, 2013; Piotroski, 2000), confirming that the model is exploiting established economic relationships rather than noise.

A secondary finding concerns prediction timing. Restricting the evaluation to stocks at business day 1 after their earnings release consistently improves out-of-sample performance relative to later days. This is consistent with the post-earnings announcement drift literature (Ball and Brown, 1968; Bernard and Thomas,

1989), which documents a gradual market adjustment to fundamental information in the period following the release date.

6. LIMITATIONS AND FUTURE DIRECTIONS

Regime instability. The most consequential open challenge is the instability of cross-sectional relationships across regimes. During crises, high within-period stock correlations amplify portfolio volatility and reduce the benefit of diversification. Future work should explore ensemble approaches combining regime-local and global models, or Bayesian updating frameworks that allow the model's effective weighting to shift as regime evidence accumulates.

Loss function alignment. The current objective—squared error on log-returns—weights all observations equally. From a portfolio perspective, only the extremes of the prediction distribution matter. A ranking-oriented loss function could better concentrate predictive power at the decision boundary.

Hypothesis testing over optimization. The current framework is oriented toward predictive performance. A complementary direction is to test specific economic hypotheses—for example, whether the earnings yield signal is driven by growth expectations or discount rate changes— which would strengthen the economic foundation and potentially identify more robust features.

7. CONCLUSIONS

This paper demonstrates that point-in-time accounting fundamental data contain robust and economically interpretable predictive information about near-term equity returns, evaluated on a strict out-of-sample test period from 2012 to 2023. The central contribution is methodological: a three-block walk-forward TSCV design that eliminates indirect temporal leakage in model selection, embedded in a fully automated end-to-end quantitative pipeline using XGBoost as the core model.

The results suggest that the predictability documented in the academic literature is not merely a statistical artefact of evaluation methodology. With appropriate attention to survivorship bias, look-ahead contamination, and evaluation design, machine learning models trained on fundamental data can generate meaningful, robust, and economically interpretable return signals over a broad range of market conditions. The main open questions—regime stability, transaction cost scalability, and loss function alignment—represent productive directions for future development.

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